

Mohsin Lone

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Python Backend Engineer with experience in designing scalable systems and data-driven applications. Proven ability to build high-performance APIs, optimize backend processes, and architect distributed systems using technologies like Redis, PostgreSQL, and Azure Event Hubs.

Experienced in building **AI-powered backend systems**, including **RAG-based search, text summarization, and automated content generation**, with a strong focus on problem-solving, system design, and performance engineering. Adept at collaborating across teams and independently owning end-to-end project delivery.

Experience

MAY 2023 – PRESENT

SDE-1 | Nyalazone Solutions Pvt Ltd | Noida, India

- Architected distributed, event-driven systems using Flask, Redis, PostgreSQL, and Azure Event Hubs, enabling reliable **real-time data processing at scale**.
- Built **AI-powered features** including **RAG-based search, text summarization, and automated email/content generation**, improving information retrieval and reducing manual effort.
- Developed **Retrieval-Augmented Generation (RAG) pipelines** by combining vector-based retrieval with LLM responses for contextual query answering over structured and unstructured data.
- Integrated **Microsoft Graph API** for email ingestion, processing, and intelligent response workflows, enabling automation of communication pipelines.
- Implemented document classification using rule-based and logic-driven approaches, improving accuracy and efficiency in handling large-scale data.
- Designed and developed a contextual chatbot system inspired by NLP and intent recognition concepts, enabling dynamic response generation and improved user engagement.
- Implemented vector-based retrieval using embeddings for semantic search over large datasets.

SEP 2022 – MARCH 2023

Teaching Assistant | Function Up | Remote

- Facilitated our juniors' understanding of technologies by serving as a TA at a coding bootcamp
- Fostered increased self-assurance and improved comprehension of acquired knowledge.

Skills

Python, TypeScript, JavaScript, Data Processing, ML Concepts, NLP (basic), Model Logic Design, Flask, Node.js, PostgreSQL, Redis, Azure Event Hubs, Graph API, System Design, Performance Optimization, Scalable Architecture, RAG, AI Search, LLM,

Education

AUG 2018 – JUL 2022

Bachelor of Technology CSE | Lovely Professional University | Jalandhar, India

- Specialized in DSA, DBMS, and Software Engineering, with hands-on experience in building scalable applications.
- Submitted a Research paper on [Self Learning Chatbots using Reinforcement Learning](#).

Projects

- **Meeting Summarizer (Integrated Teams with CRM System):** Used Microsoft Graph API to integrate Microsoft Teams with a CRM system, generating **AI-based meeting summaries and insights**. Implemented **RAG-based contextual retrieval** over transcripts to improve information access. Improved note-taking efficiency by **10x**, reduced errors, and enabled **automated task creation** from discussions.
- **Email Polling System:** Built an email polling microservice using Azure Event Hubs, Redis, and Celery, integrating Microsoft Graph API with AI-based classification, summarization, and generation, improving efficiency by **30%** and reducing latency by **25%**.